FLATNESS PLUS FIBRE DEPTH AT LEAST ONE CUTS OUT A SLICE THAT STAYS FLAT AND FINITELY PRESENTED

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 6336-6349 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 6351-6355, the complete author proof (3 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. The cited lemma-slice-given-element itself failed legacy mechanical revalidation (one of the four known old-wave FAILs) and is not registered as its own card; the dependency is noted here for the reader.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The slice lemma for a prescribed element (lemma-slice-given-element, same chapter) which turns a fibre-nonzerodivisor into a flat finitely presented Cartier divisor; cited input with the choice of h written in the retained span.

Source reference. Original lemma 056Y: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6166–6177. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Let $x \in X$ be a point with image $s \in S$. Assume*

(1) *f is locally of finite presentation,*
(2) *f is flat at x, and*
(3) *$\mathcal{O}_{X_s,x}$ has depth ≥ 1.*

Then there exists an affine open neighbourhood $U \subset X$ of x and an effective Cartier divisor $D \subset U$ containing x such that $D \to S$ is flat and of finite presentation.

Proof. Pick any $h \in \mathfrak{m}_x \subset \mathcal{O}_{X,x}$ which maps to a nonzerodivisor in $\mathcal{O}_{X_s,x}$ and apply Lemma 056Y. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.